

# Automated Occupancy Detecting AC Control System

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**Abstract**—Energy wastage due to unattended air conditioning systems is a major problem in residential and commercial buildings. This project proposes an automated occupancy detecting air conditioning control system by using PIR and microwave sensors integrated with an IoT controller. The system applies energy-saving control strategies based on occupancy duration.

**Index Terms**—Occupancy detection, air conditioning control, energy efficiency, PIR sensor, IoT

## I. INTRODUCTION

Air conditioning is a big global energy consumer. Much of the power is wasted simply because people forget to turn Air conditioning units off. This negligence carries a heavy cost.

In Sri Lanka, where electricity generation is highly dependent on non-renewable sources such as coal and fuel oil, this waste directly contributes to increased greenhouse gas emissions and global warming. As well as the continuous and unnecessary operation of these units places undue stress on critical components like compressors and fan motors, leading to reduced equipment lifespan and higher maintenance costs.

An easy approach to automation—frequently turning the AC on and off based on movement, creates a secondary engineering problem called short cycling. Rapid power cycling causes high wear on the compressor and may ultimately consume more energy than continuous operation. Therefore an effective solution must distinguish between short-term absences (as a example user leaving for a break) and long-term unoccupancy.

This project proposes the design of an Automatic Occupancy Detecting AC Control System used to minimize energy waste while protecting AC equipment. The system utilizes a sensor fusion approach, combining Passive Infrared called as PIR and Microwave sensors to detect human presence with higher accuracy than single sensor systems.

The primary objective of this system is to bridge the gap between human negligence and energy efficiency in tropical environments utilizing split-type AC units. By automating the control process and integrating features such as IoT connectivity and manual QR code overrides, the project aims to provide a practical, user-centric solution that reduces electricity bills and environmental impact without compromising user comfort.

## II. LITERATURE REVIEW

Research consistently shows that automating HVAC systems based on occupancy is a highly effective way to cut down on energy waste, largely because it addresses the common human habit of forgetting to turn off the air conditioner. [1]. By integrating accurate occupancy data with building control systems, buildings can reduce unnecessary heating, cooling, or lighting, leading to significant energy savings. [2] Standard sensors often struggle to detect people when they are sitting still like when reading or sleeping which causes the AC to turn off incorrectly. To solve this, modern strategies suggest sensor fusion, which combines different inputs like motion and Carbon Dioxide levels to accurately confirm whether someone is actually in the room. Crucially, the system also needs to be gentle on the hardware, simply switching the AC on and off frequently causes short cycling, which wears out the compressor and shortens the unit's lifespan. Therefore, a truly smart solution must distinguish between a quick break and a long absence, using standby modes to protect the equipment—a feature often missing in the split-type AC units common in tropical regions like Sri Lanka [3]

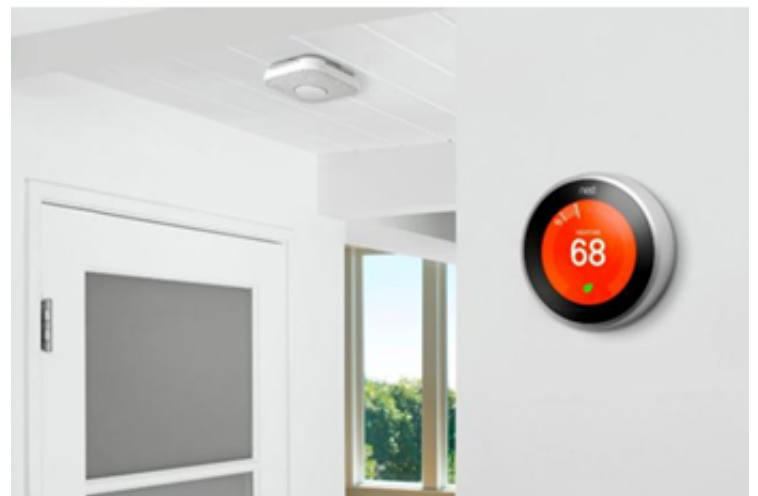


Fig. 1: System Design of the automated occupancy detecting AC control system

### III. PROBLEM STATEMENT

A major reason for energy waste in modern buildings is the simple habit of leaving air conditioners running in unoccupied rooms. Because most standard AC units lack the automation to detect whether someone is actually present, they continue to operate at full capacity even when cooling empty spaces. This negligence carries a surprisingly high cost, for instance, a single unit (18000 BTU) left unattended can waste between 10 to 14 kWh of electricity in just eight hours. [4] When this issue is multiplied across large offices or industrial facilities, it creates a massive, avoidable drain on energy resources that manual control alone often fails to prevent.

The consequences of this continuous operation extend far beyond inflated electricity bills. In Sri Lanka, where the power grid relies heavily on non-renewable sources like coal and fuel oil, this wastage directly contributes to increased greenhouse gas emissions and global warming. Furthermore, running the system unnecessarily places constant stress on critical components like the compressor and fan motor. This accelerates wear and tear, significantly reducing the equipment's lifespan and leading to frequent, expensive maintenance issues that could otherwise be avoided. [5], [6]

### IV. CONTROL LOGIC FLOWCHART

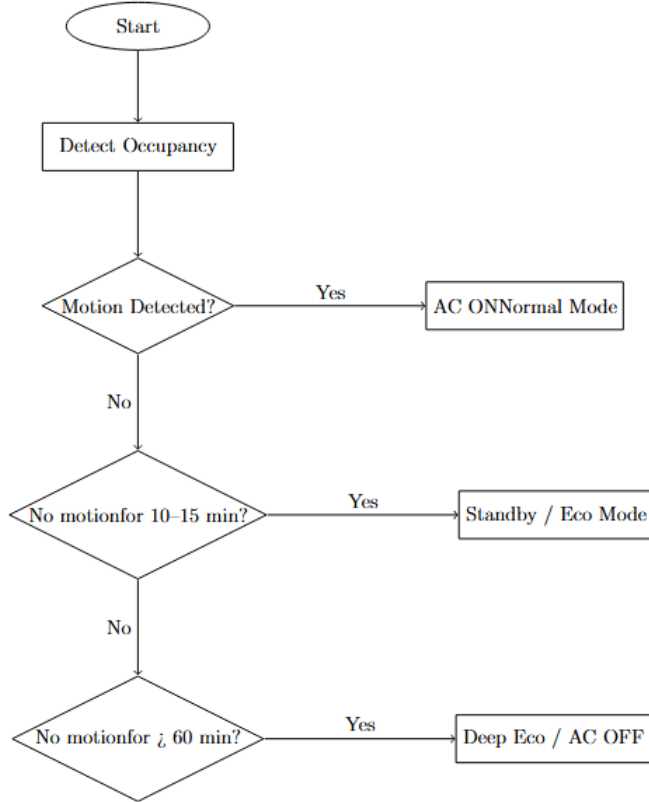


Fig. 2: Control Logic Flowchart

The control algorithm continuously monitors occupancy using motion sensors. If motion is detected, the air conditioner

operates in normal mode. When no motion is detected for a short duration (10–15 minutes), the system switches the AC to standby or eco mode to save energy while maintaining comfort. If the room remains unoccupied for a long duration exceeding 60 minutes, the controller switches the AC to deep eco mode or turns it off completely. This staged control prevents frequent on/off cycling and protects the compressor.

### V. SYSTEM OVERVIEW

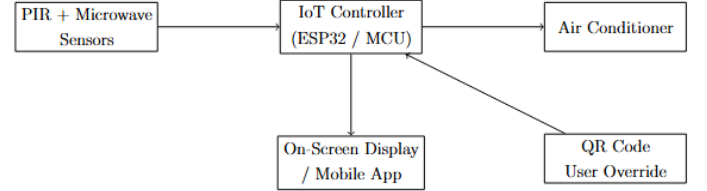


Fig. 3: Overall system Design

The system consists of occupancy sensors, an IoT-based controller, and an air conditioning unit. PIR and Microwave sensors detect the presence of humans and send signals to the controller. Based on the detected occupancy duration, the controller determines the appropriate air conditioning operating mode. An optional display or mobile application provides real time system information, while a QR-code-based override allows manual user interaction. The system consists of occupancy sensors, an IoT-based control unit, and an optional user interface. The controller incorporates timers and anti-short-cycling logic to protect the compressor from frequent ON/OFF operation.

### VI. SYSTEM DESIGN

When designing an automated system to reduce energy wastage from unattended air conditioners, the major concern with having it turn on and off frequently is the wear and stress it causes to the equipment of the air conditioning unit (especially the compressor). In fact, turning it on and off frequently may consume more energy than keeping it running for a short period. Therefore, we must design a system which distinguishes between short-term absences and long term unoccupancy.

Modern air conditioners are equipped with a Standby or Eco mode. This is a low power mode which slows down the compressor and increases the temperature by 2-3 °C. When the PIR + Microwave sensors do not detect movement in the room for around 10-15 minutes the IOT based control unit would send the signal to the air conditioner indicating a short-term absence. The air conditioner then switches into Standby / Eco mode instead of turning off. Therefore, when the room becomes occupied again the compressor wouldn't have to start from a fully powered-off state, which reduces wear and prolongs the lifespan of the AC unit. [7], [8] Furthermore, if the sensors do not detect movement for more than 60 minutes the IOT based control system would send the signal to the AC

indicating a long-term absence. The AC would then either turn off completely or switch to Deep Eco mode (If it has one).

The sensors should be installed in a position such that they are facing the main activity areas of the room, such as the office desk or sitting areas. Many sensors nowadays have 90° – 180° detection angle allowing them to cover most of the room. However, they should be installed away from the direct air flow of the AC. The QR code is provided for the user in case of a need to manually override the automatic system. The user can manually override the system to turn off the AC (bypassing the Eco mode sequence) if he is leaving for an extended period.

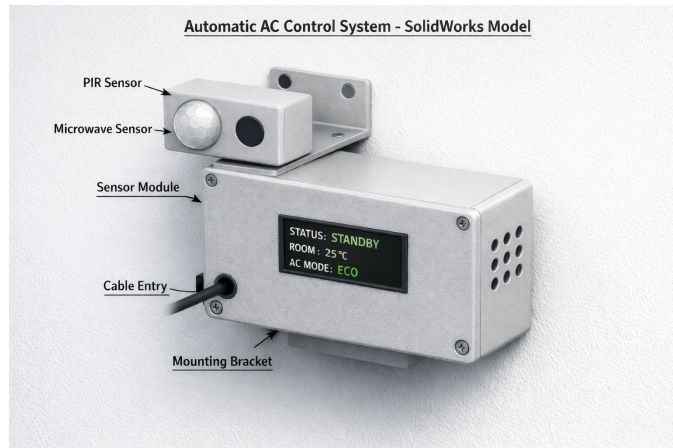


Fig. 4: system design of the AC control system

#### I. PIR + Microwave Sensors for occupancy detection

- Function – Detects human presence in real time.
- PIR (Passive Infrared) sensors detect changes in infrared radiation emitted by human bodies.
- Microwave sensors detect motion by sending waves and measuring reflections.
- Both sensors require movement to trigger the sensor.

#### II. AC Control Unit (IOT based)

- Function – Receives occupancy data from sensors and controls the AC.
- Receives signals from the PIR + microwave sensors.
- Switches AC between normal, standby/eco mode, and full off/deep eco mode based on occupancy.
- Ensures safe operation by preventing rapid on/off cycles.

#### III. On-Screen Display (Optional)

- Function – Displays real-time information of the status in the room.
- Information such as occupancy status, AC mode (ON/Standby/OFF), room temperature and AC temperature can be displayed.
- Help identify system faults.

#### IV. Standby / Eco Mode Logic

- Function – Saves energy when the room is unoccupied.

- After 10–15 minutes of no detected motion AC goes into standby/eco mode.
- Compressor slows down, temperature setpoint raised slightly (+2 – 3 °C).
- Maintains comfort if occupants are stationary.

#### V. Deep Eco Mode / Turn Off Logic

- Function – Saves maximum energy during long absences.
- When no motion is detected for more than 60 minutes the AC fully shuts down or goes into Deep Eco Mode (If it has one).
- The sensors remain active and the AC resumes automatically when motion is detected.

#### VI. Timers and IoT Controller Logic

- Function – Central brain of the system coordinating with sensors, AC modes, and timing.
- The timer tracks periods of unoccupancy and prevents short cycling of the AC which protects the equipment.
- Optional – Can integrate with a mobile app to display real-time room status. (Occupancy, AC mode, Room temperature etc.)

#### VII. Optional User Override / QR Code

- Function – Allows occupants to manually override the automatic system using a QR code.
- Each occupant is provided with a QR code (Can be printed in the person's employee card).
- The occupant can scan it upon entering, to let the system know that the occupant is here and when leaving for long absences, that the room will be unoccupied for a while.
- This gives the user a fair amount of control over the automatic system.

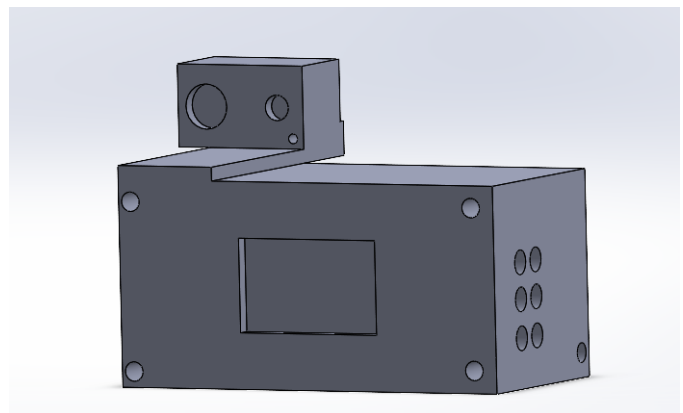


Fig. 5: Solid Work Design of the Automatic Occupancy Detecting AC control system

The mechanical housing for our AC control system was designed in SolidWorks using a practical, three part assembly consisting of a main control box, a dedicated sensor unit, and mounting brackets. We prioritized functionality and airflow, designing the main enclosure with side ventilation slots to

prevent the internal electronics from overheating, along with a custom cutout on the front face to perfectly fit the system display. To ensure the best possible performance for occupancy detection, the PIR and microwave sensors were housed in a separate upper unit; this elevation allows them a better field of view across the room. We modeled each of these components individually and then combined them in a virtual assembly to verify that all parts fit together correctly. The final step of the design involves installing the physical components, securing the display screen into the front slot and connecting the sensors in the top housing to create a robust, fully integrated control device.

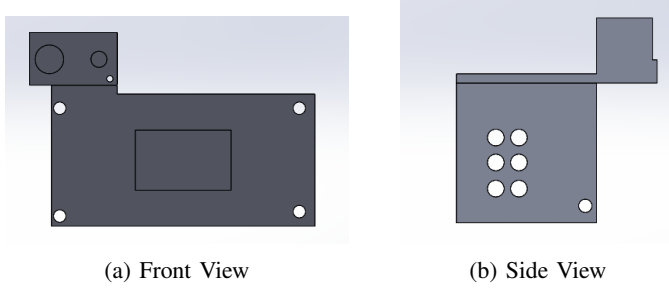


Fig. 6: SolidWorks design views of the AC Control System enclosure.

## VII. MATLAB MODELS AND SIMULATION RESULTS

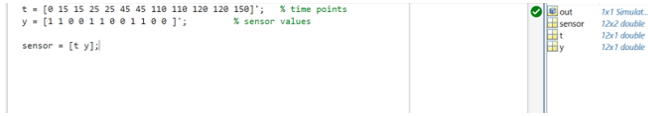


Fig. 7: MATLAB code for the sensor inputs

The MATLAB simulation is prepared to validate the responsiveness of the control logic by creating a synthetic input signal that mimics the behavior of the physical sensor module. In this setup, the occupancy status of the room is represented as a simple binary value, where a signal of 1 indicates that the room is occupied and a signal of 0 indicates that it is currently empty. This binary approach allows the system to be tested against precise, deterministic patterns of movement and absence without the noise or variability of real-world sensor data.

A critical aspect of this simulation is the use of time scaling to make the testing process efficient. Because the control system is designed to react to long periods of inactivity—specifically waiting up to an hour before initiating a full shutdown—testing these features in real-time would be impractical and time-consuming. To address this, the simulation scales time so that one second of simulation corresponds to one full minute of real-world operation. Consequently, the total simulation duration of 150 seconds effectively represents a monitoring window of 150 minutes, or two and a half hours. This duration is sufficient to verify the entire range of the

system's logic, including the initial fifteen minute trigger for the standby mode and the sixty minute threshold for the deep energy-saving mode, all within a rapid testing timeframe.

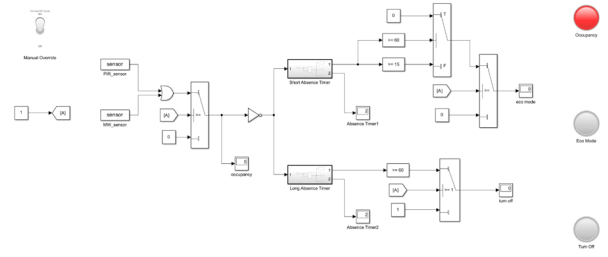


Fig. 8: Simulink for the Automated Occupancy Detecting AC Control System

This Simulink model implements an occupancy-based energy-saving control system using motion sensors and timed absence detection. The system primarily relies on a Passive Infrared (PIR) sensor and a Microwave (MW) sensor to detect human presence. The outputs of the two sensors are combined for more accurate detection. A manual override switch is included, allowing the user to force the system on or off regardless of the sensor outputs.

When occupancy is not detected, it activates the two absence timers. One is the short-term absence detecting timer and the other is the long-term absence detecting timer. When the short absence timer hits 15 (indicating 15 minutes of unoccupancy) the system goes into Eco Mode (the Eco mode display goes to 1). When both timers hit 60 (indicating 60 minutes of unoccupancy) the system shuts down (Eco Mode turns off too). From-to blocks have been implemented to send the signal of the manual override switch, allowing the user to turn on/off the whole system at any time.

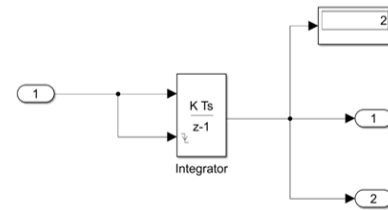


Fig. 9: Short / Long Absence Timer Subsystems

The Short and Long Absence Timer subsystems function as the precise internal clocks of the control logic, responsible for measuring exactly how long a room has been left empty. Rather than using simple mechanical timers, the system implements this digitally using discrete-time integrator blocks within the simulation. These integrators act effectively as accumulating counters; as long as the sensors report that the room is unoccupied, the blocks continuously add up the time,



second by second, creating a running total of the duration of absence.

Crucially, this counting mechanism is designed to be highly sensitive to human presence to prevent false shutoffs. The moment the sensors detect even the slightest movement—indicating that the room is occupied—the input to the timer subsystems changes, causing the integrators to instantly reset to zero. This “hard reset” feature ensures that the countdown doesn’t just pause but restarts completely every time a person moves. This guarantees that the energy-saving modes (Eco or Shutdown) are only triggered after a continuous, uninterrupted period of vacancy, ensuring that users are never plunged into an uncomfortable environment due to a cumulative timing error.



## VIII. CONCLUSION

The Automatic Occupancy Detecting AC Control System offers a practical and robust solution to the widespread problem of energy wastage caused by leaving air conditioners running in empty rooms. By moving beyond simple motion detection and adopting a sensor fusion strategy and combining PIR and Microwave technologies, the design effectively solves the common problem of false shutoffs, ensuring that the system remains active even when occupants are sitting quietly. Crucially, the system is engineered to protect the hardware itself. Rather than immediate cutting power which causes damaging short cycles, the intelligent staged logic transitions to an energy saving Eco Mode for short breaks and only starts a full shutdown after prolonged absence. This balanced approach successfully delivers significant electricity savings and reduces environmental impact, all while preserving user comfort and extending the operational lifespan of the air conditioning unit.

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